

Rover Rescue: Plan a Fair Test

GROW • Science • 40 minutes

I can plan a fair investigation of how surface type affects a rover's travel distance.

Prepare before pupils arrive

Setup: Prepare a stable low ramp, one rover, smooth and fabric lanes, a marked start, ruler and six-row table.

Safety: Teacher approves the lane and catches runaway equipment. Use a director or digital route for mobility, sensory or handling access.

Equipment: Planning mat; low ramp; trolley or rover; smooth and fabric surfaces; metre rule or tape; start and finish markers.

Safety: Teacher checks a stable low ramp and clear lane. Do not push the rover, chase it or place hands in its path.

Fallback: Plan with cards only or direct an adult to place every method card. No equipment handling is required.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Consolidate and revisit key ideas through investigation.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 4 min	Name the independent variable in a surface test. Name the dependent variable and its unit.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter · 4 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.
I do · 5 min	independent variable — what I change dependent variable — what I measure control variable — what I keep the same Choose one question: How does surface type affect rover travel distance? Conditions: smooth and fabric.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.

Stage / total time	Teaching move	Adult support / response
We do · 5 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 4 min	Safety first: use a stable low ramp in a clear tray or lane. Release without pushing and keep hands out of the travel path.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Place each method card under Change, Measure or Keep the same. Reject any plan that changes two variables.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 10 min	Produce a safe fair-test plan comparing smooth and fabric surfaces, with three repeats per condition. Record: Question, prediction, variable table, safe numbered method, two conditions, six result spaces and one model limitation.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Name the dependent variable and two controls. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Choose one visible next move. The pupil writes what they said and what changed.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: How does surface type affect travel distance when ramp height and release stay the same?

Yes. It changes one independent variable, measures distance and names important controls.

Misconception repair

Repaired. A fair comparative test changes one independent variable while measuring one outcome and controlling the rest.

Guided checkpoint

Yes. It changes one independent variable, measures distance and names important controls. Choose the question with one deliberate change and a measurable outcome.

Surface type

CHANGE. This is the chosen independent variable. Ramp height must now stay fixed. For this question, which one factor is deliberately different?

Travel distance in cm

MEASURE. This is the dependent variable and includes a clear unit and measuring rule. Which card describes the result collected after each run?

Low ramp height

KEEP THE SAME. Ramp height is controlled so surface is the only deliberate change. If surface is the independent variable, ramp height must remain constant.

Same rover

KEEP THE SAME. Using the same rover removes a second possible cause of distance differences. Changing the vehicle would make the surface comparison harder to interpret.

Same no-push release

KEEP THE SAME. A consistent release helps make the runs comparable. A different push would add another changing force.

Three results per condition

REPEAT / RECORD. Repeats reveal variation and give a more dependable pattern than one run. Repeats are part of the reliability and recording plan, not the dependent variable.

Model limitation

The classroom rover, surfaces and short ramp simplify a real planetary rover. The model can compare friction here, not copy every real condition.

Independent evidence

Question, prediction, variable table, safe numbered method, two conditions, six result spaces and one model limitation.

Supported: Use the prepared surface question and complete Change, Measure, Keep and Repeat with icon-backed cards. Adult places equipment only as directed; a six-row table and unit are supplied.

Standard: Write a fair question, prediction, variable table and numbered method for two conditions with three repeats. Use the check: one change, one measured outcome, same release and safe low ramp.

Stretch: Justify the controls, predict random variation and explain one limitation of transfer to a real rover. Optionally suggest a sensible condition order without changing the safety rule.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: [SCI_G_W13A_Rover_Rescue_Plan_Explore.html](#)

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.